Phishing Detection Tool with Real-Time Threat Intelligence

ILLINOIS INSTITUTE
OF TECHNOLOGY

Project Management Plan

Department of Information Technology and Management

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Version 2.0

Revision History

Note: The revision history cycle begins once changes or enhancements are requested after the document has been baselined.

Date	Version	Description	Author
11/27/2024	2.0	Final Project Management Plan	Madhuri Kethineni
11/12/2024	1.0	Initial Project Management Plan Draft	Madhuri Kethineni

Artifact Rationale

The Project Management Plan (PMP) acts as a source document of the "Phishing Detection Tool with Real-Time Threat Intelligence" to warrant the achievement of the intended project. It offers a framework of managing project by planning out the course of action of every activity so that objectives, timeframes and product targets are considered. The PMP includes key planning assumptions, decision, and control limits for scope, cost and schedule, so it provides direction for the project team and stakeholders.

Since the promotional of accountability and consistency throughout the stages of project life cycle, the PMP enhance communication and documentation of responsibilities, risk management and quality assurance. It can be easily modified and updated to address changes in project requirements to restore original project objectives and defining factors for success.

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1. Introduction

This PMP (Project Management Plan) describes the project management processes that will follow during execution of the "Phishing Detection Tool with Real-Time Threat Intelligence" project. The project's processes will align with plans and processes of the Project Management Accountability System (PMAS) guide new processes will be defined for any management areas not covered by the PMAS Guide. This PMP will govern the management practices across the life of the project.

1.1 Project Overview

This is the "Phishing Detection Tool with Real-Time Threat Intelligence" which seeks to design a smart program that can detect phishing threats that are real-time and the data from four different APIs. There will be graphical user interface (GUI) of the tool that takes threat intelligence and processes them for easy understanding to fight off phishing attacks.

There are three main sections in the project: design, implementation, and testing; and each phase took one week to be completed. "**Python**" will be used for back-end processing, '**Tkinter'** framework will be used for GUI, and "**VirusTotal, Abuse IPDB, IPQS and IP info**" APIs for threat visibility. The result of the process will be the OSS, presented in the GitHub, which develops an opportunity of real-time detection of threats as a response to the emergent cybersecurity issues.

1.2 Scope Statements

- Proposal for the creation of a Phishing detection tool.
- Inclusion of at least four APIs/data source for real time threat intelligence.
- A graphical user interface for communication or data presentation.
- Establishing a GitHub repository to use versioning for all code and documentation.

1.3 Goals and Objectives

- To develop a useful, simple and open-source tool that may navigate to phishing activities with real time threat intelligence.

- Make sure they must integrate four APIs for receiving real-time updates.
- Develop a solution that detects phishing with accuracy.

1.4 Stakeholders and Key Personnel

Stakeholders and key personnel for this project are:

- Project Manager: Madhuri
- Software Developers: Vijay, Mahesh
- Quality Assurance Engineer: Branden, Radha
- Technical Writer: Usha Sree
- End-Users: Business Owners, Educational Institutes

2. Project Organization

- Using “Agile methodology” with a week as the duration of a sprint.
- Feature branches should be used for code development while merging back to the main branch is done testing is completed.
- Perform integration tests on all the distinct parts of unit.
- Perform user acceptance testing (UAT) before deployment is done.
- Detailed documentation for each development phase will be kept in the GitHub repository.

3. Acquisition Process

- The project team utilize open-source resources and tools.
- Tools: Python IDE, Tkinter for GUI, APIs (VirusTotal, Abuse IPDB, IPQS, IP info).

4. Monitoring and Control Mechanisms

This project follows standard monitoring and control processes as defined for risk management, requirements traceability, and operational readiness.

5. Systems Security Plans and Requirements

As a part of the project's planning phase, the 'Phishing Detection Tool with Real-Time Threat Intelligence' follows best practices for security and privacy.

6. Work Breakdown Structure (WBS) and Schedule

Phase	Deliverables	Timeline
Sprint 1	Project setup, API research, and basic GUI	Week 1
Sprint 2	Integrate APIs, improve GUI, initial testing	Week 2
Sprint 3	Development, testing, documentation	Week 3
Sprint 4	Final testing, deployment, and presentation	Week 4

7. Project Success Criteria

The Phishing Detection Tool with Real-Time Threat Intelligence is developed and deployed and meet the requirements.

8. Communication Management Plan

Communication Strategies:

- Channels: WhatsApp, Google Meet, Emails
- Shared the documentation on Google Drive.

Collaboration Tools:

- GitHub for handling code and for versioning.
- Pull requests for the purpose of peer reviews on GitHub

9. Risk Management Plan

- Use backup APIs.
- Usability testing as often as possible.
- Ensure that time is attached to important and sensitive activities.

10. Software Configuration Management (SCM) Plan

- GitHub repository version controls.
- Weekly code reviews.

11. Training Plan

- Teach all team members about the API and about how to create a GUI.
- Training members are development team.
- Teach them in week 1 and week 2.

12. Quality Assurance Plan

- Unit test, integration test, performance test, penetration test and GUI usability test.
- Cover all basic facilities and all QA activities should be recorded in GitHub.
- QA phases will then correspond with the project deliverable phase.

13. Project Measurement Plan

The plan creates awareness of the goals and associated performance metrics to make certain that the project remains focused to achieve its goals.

13.1 Description

Measurement Objectives: Effectiveness of timely task completion for each assigned task, effectiveness of the phishing detection tool, and efficiency of threat detection.

Metrics:

- Task progress: Of these tasks, the percentage that was completed on time.
- System functionality: Total number of APIs that were integrated successfully into the system.
- Detection accuracy: Detection rate of phishing against benchmark datasets.

- Data Collection: In simple words, it is more effective to use GitHub for committing the tasks and, in general, to test the performance of the system.
- Reporting: Weekly updates about the project.

13.2 Performance Measurements

Project “Phishing Detection Tool with Real-Time Threat Intelligence” performance measurements is as follows:

No.	Measurement Name	Measurement Objective	Metric
1.	Task Completion	Work in progress against goals & targets	80% of the total tasks accomplished on time
2.	API Integration	Ensure all APIs are working	APIs integrated successfully
3.	Detection Accuracy	Accurate the performance of the detection of phishing	90% accuracy in detection tests
4.	Usability Testing	Make GUI as responsive as your target user expects	Satisfaction of user

14. Reference Materials:

- AbuseIPDB. (n.d.). *Check the report history of an IP address* https://www.abuseipdb.com/
- IPinfo.io. (n.d.). *IP address details and API*. https://ipinfo.io/
- IPQualityScore. *Fraud prevention & detection tools*. https://www.ipqualityscore.com/
- VirusTotal. (n.d.). *Analyze suspicious files and URLs*. https://www.virustotal.com/
- GitHub Collaboration Guide.
- Project Management Plan Template provided by IIT.